

Optimal Order Placement Strategy for Liquidity Traders

Report on Market Microstructure & Execution Probability

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Abstract

This report synthesizes the theoretical and empirical frameworks for optimal order placement strategies, specifically focusing on "Liquidity Traders" who face a hard execution deadline. By utilizing a Mean-Variance Utility Maximization framework, we balance the trade-off between the superior pricing of limit orders and the risk of non-execution. The report integrates fundamental definitions from market microstructure with advanced statistical models, including Arithmetic Brownian Motion, Empirical Density Estimation, and Conditional models using ACD and ARMA-GARCH frameworks.

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1 Market Microstructure Fundamentals

Based on the foundational notes on market dynamics , we define the state of the market through the Limit Order Book.

1.1 Order Book Definitions

Let $x_p(t)$ denote the number of unexecuted limit orders at price level p at time t . The state of the order book $X(t)$ is defined by the set of all $x_p(t)$.

- **Best Ask Price ($P_A(t)$):** The lowest price at which a seller is willing to trade.

$$P_A(t) = \inf\{p : x_p(t) > 0\} \quad (\text{Sell Orders}) \quad (1)$$

- **Best Bid Price ($P_B(t)$):** The highest price at which a buyer is willing to trade.

$$P_B(t) = \sup\{p : x_p(t) > 0\} \quad (\text{Buy Orders}) \quad (2)$$

- **Bid-Ask Spread ($S(t)$):** Represents the cost of immediacy.

$$S(t) = P_A(t) - P_B(t) \quad (3)$$

- **Mid-Price ($P_m(t)$):**

$$P_m(t) = \frac{P_A(t) + P_B(t)}{2} \quad (4)$$

1.2 Agent Types

Traders are categorized by their urgency :

1. **Impatient Agents:** Value immediate execution over price improvement. They typically utilize Market Orders.
2. **Patient Agents:** Value price improvement over immediacy. They typically utilize Limit Orders and bear execution risk.

2 The Liquidity Trader's Problem

We consider a Liquidity Trader who must purchase a set quantity of an asset before a strict deadline T . The trader faces three primary execution choices:

1. **Immediate Execution:** Submit a market order at $t = 0$ at price p_0^M .
2. **Delayed Execution:** Wait and submit a market order at $t = T$ at price p_T^M .
3. **Limit Order Strategy:** Submit a limit order at $p^L < p_0^M$. If the order is not executed by time T , the trader must aggressively cross the spread and buy at the closing market price p_T^M .

3 Mathematical Framework

To determine the optimal limit price p^L , we employ a Mean-Variance Optimization framework .

3.1 Payoff Function

The utility $U(p^L)$ is a random variable dependent on the execution state:

$$U(p^L) = \begin{cases} U_E(p^L) & \text{if executed (Success)} \\ U_{NE}(p_T) & \text{if not executed (Failure)} \end{cases} \quad (5)$$

Where $U_E = p_0^M - p^L$ represents the savings (profit) from the limit order, and $U_{NE} = p_0^M - p_T^M$ represents the cost or gain relative to the initial price if forced to trade at the deadline.

3.2 Expected Utility and Variance

Let $P_E(p^L)$ be the probability of execution. The expected utility is defined as:

$$E[U(p^L)] = P_E(p^L)U_E(p^L) + [1 - P_E(p^L)]E[U_{NE}|\text{No Exec}] \quad (6)$$

The variance of the utility, representing risk, is:

$$V[U(p^L)] = P_E(1 - P_E)(U_E - E[U_{NE}|\text{No Exec}])^2 + (1 - P_E)V[U_{NE}|\text{No Exec}] \quad (7)$$

3.3 Optimization Objective

The trader selects p^L to maximize the risk-adjusted return:

$$\hat{p}^L = \arg \max_{p^L} (E[U(p^L)] - \lambda V[U(p^L)]) \quad (8)$$

where λ is the risk aversion parameter .

4 Modeling Execution Probability and Risk

To solve the optimization problem, we model P_E and the conditional distribution of p_T .

4.1 Theoretical Model: Arithmetic Brownian Motion

Assuming price follows $dp_t = \mu dt + \sigma dW_t$, the probability of execution is derived from the First Passage Time (FPT) distribution :

$$P_E(p^L|p_0) = \Phi\left(\frac{p^L - p_0 - \mu T}{\sigma\sqrt{T}}\right) + e^{\frac{2\mu(p^L - p_0)}{\sigma^2}} \Phi\left(\frac{p^L - p_0 + \mu T}{\sigma\sqrt{T}}\right) \quad (9)$$

4.2 Empirical Unconditional Model

This model utilizes historical price fluctuations M_T^B (maximum price drop) to estimate probabilities directly from data .

$$P_E(p^L) = 1 - \frac{1}{N} \sum_{i=1}^N I\{\Delta_i < p_0 - p^L\} \quad (10)$$

The non-execution price is estimated using a **Left-Truncated Distribution**, considering only historical periods where the price did *not* drop low enough to trigger the limit order.

4.3 Empirical Conditional Model (ACD + ARMA-GARCH)

This dynamic model adapts to current market conditions (volatility clustering) .

4.3.1 1. Execution Probability: ACD Model

The **Autoregressive Conditional Duration (ACD)** model predicts the expected price fluctuation ψ_i . The ABAMACD specification updates expectations based on recent shocks:

$$\psi_i^{\delta_1} = \omega + \sum \alpha_j(\dots) + \sum \beta_j \psi_{i-j}^{\delta_1} \quad (11)$$

Higher ψ_i implies higher market speed and higher probability of execution.

4.3.2 2. Non-Execution Risk: ARMA-GARCH

The **ARMA-GARCH** model forecasts the trend (μ) and volatility (σ) of the closing price. Crucially, we use a **Truncated Mean** to account for adverse selection:

$$E[p_T|\text{No Exec}] = \mu + \sigma \times \left(\frac{\phi(z)}{1 - \Phi(z)} \right) \quad (12)$$

where $\frac{\phi(z)}{1 - \Phi(z)}$ is the Inverse Mills Ratio.

4.4 Maximum Likelihood Estimators (MLE)

To implement the conditional models, parameters θ are estimated via MLE.

A. MLE for ACD (Execution Probability) :

$$\hat{\theta}_{ACD} = \arg \max_{\theta} \sum_{i=p+1}^N \ln \left(F_{GG} \left(\frac{\Delta_i + \delta}{\psi_i s(t_i)} \right) - F_{GG} \left(\frac{\Delta_i}{\psi_i s(t_i)} \right) \right) \quad (13)$$

where F_{GG} is the Generalized Gamma CDF and $s(t_i)$ is the seasonality component.

B. MLE for ARMA-GARCH (Price Risk) :

$$\hat{\theta}_{ARMA} = \arg \max_{\theta} \sum_{i=p+1}^N \left[\ln f_{NN} \left(\frac{z_i}{\sigma_i} \right) - \ln \sigma_i \right] \quad (14)$$

where z_i is the residual error and f_{NN} is the standard Normal PDF.

5 Summary of Results

The research indicates that dynamic strategies (Conditional Models) typically outperform static strategies by adapting p^L to market volatility. The truncated distribution significantly improves the estimation of non-execution cost by correctly accounting for the "conditional" nature of failed trades.